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MILITARY SPECIFICATION

PENDULUM

(For Torpedo MK 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement, inspection, and test of a pendulum, LD 296723.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-C-24308	Connector (Part Number M24308/3-2)
MIL-E-82590	Environmental Requirements and Tests (For Torpedo MK 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base

FSC 1355

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DRAWINGS

Naval Ordnance Systems Command

LD 296723	Pendulum, and all parts and subassemblies listed thereon
LD 495859	Shipping (Packing) Condition For One Pendulum Assembled
Dwg 677036	Pinion Wrench
Dwg 859855	Pendulum
Dwg 859875	Magnet
Dwg 877366	Pendulum Assembled

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards Publications

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Description. The pendulum consists of a framework on which is mounted a suspended pendulum that is spring restrained, magnetically damped, and actuated by gravity. Tilting of the pendulum base from the level position, in the proper plane, causes a corresponding deflection of the pendulum in relation to the base. The rotor of a synchro transmitter is connected to the pendulum and the deflection generates an ac signal which is a measure of the degree of tilt.

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3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwised specified in the contract or order, preproduction samples of the pendulum shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing pendulums that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the pendulum shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 296723.

3.2.4 Finishes, coatings, construction, and dimensions. The finishes, coatings, construction, and dimensions for the pendulum shall be strictly in accordance with applicable drawings, specifications, and publications listed in LD 296723.

3.2.5 Moisture, fungus, and corrosion. The pendulum shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications listed in LD 296723.

3.2.6 Marking. The marking of assemblies, subassemblies and parts shall be in accordance with the applicable drawings listed in LD 296723.

3.2.7 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or the appointed agent for interpretation and clarification.

3.3 Performance and product characteristics. The pendulum shall have the product characteristics and meet the performance requirements given below: (All ac voltages, unless otherwise specified, shall be rms values.)

3.3.1 Input power supply. The input supply shall be 115 volts plus or minus 3 percent, 400 Hertz (Hz) plus or minus 2 percent, single phase.

3.3.2 Voltage. When energized as specified on Drawing 877366 with

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power supply as specified in 3.3.1, the pendulum output voltage at zero degrees tilt shall be no more than 0.13 volt, and shall be repeatable within 0.05 volt as described in 4.5.4.3.1.

3.3.3 The output voltage shall vary, without sudden dips or peaks, as the pendulum base is tilted slowly in each direction.

3.3.4 The output voltage shall be at least 0.11 volt per degree tilt of the base of the pendulum over a range of base tilt from 2 degrees to 10 degrees in each direction when working into a 5000 ohms plus or minus 10 percent resistive load.

3.3.5 Phase. The output voltage shall be in phase with the supply voltage when the stand is tilted counterclockwise (as viewed from the pendulum end). When tilted clockwise, the output voltage shall be 180 degrees out of phase with the supply voltage.

3.3.6 At angles of two degrees to 90 degrees the output for equal tilt angles in each direction must be equal to within 10 percent. There shall be no discontinuities or sudden voltage changes from 90 degrees in one direction, through zero to 90 degrees in the other direction.

3.3.7 Damping. Minimum damping shall be 10 percent of critical damping. Such damping will result in successive swings to be in the amplitude ratio of 0.73 or less.

3.3.8 Movement. The pendulum shall swing freely. The pendulum shall not touch the cover nor any part of the framework under a deflection of the base of 90 degrees from the level position.

3.3.9 Natural frequency. The natural frequency of the pendulum assembly minus the Damping Magnet, Drawing 859875, shall be at least six Hertz.

3.3.10 Connector. All leads shall be brought out of the pendulum through Connector MIL-C-24308(3-2), as shown on Drawing 877366.

3.3.11 Insulation. All wiring shall withstand a hi-pot test of 500 volts, 60 Hertz for one minute, between all pins (connected together) of P1 and ground and between pins 1 and 2 connected together and pins 7 and 8 connected together.

3.3.11.1 The resistance between pins 1, 2, 7, and 8 of P1, connected together, and ground shall not be less than 20 megohms at 500 volts dc.

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3.3.12 Operating life. The operating life of the pendulum shall be not less than 500 hours.

3.4 Environmental requirements. The pendulum shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.4.

3.4.1 Vibration. The pendulum shall meet the following vibration requirements:

3.4.1.1 Vibration, nonoperating. The pendulum shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.1.2.

3.4.1.2 Vibration, operating. The pendulum shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.4.1.3.

3.4.2 Temperature and humidity

3.4.2.1 High temperature and humidity, nonoperating. The pendulum shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.1.

3.4.2.2 High temperature, operating. The pendulum shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.2.

3.4.2.3 Low temperature, nonoperating. The pendulum shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.4.2.3.

3.4.2.4 Low temperature, operating. The pendulum shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.2.4.

3.5 Cleanup. Prior to and after final assembly all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. All burrs, sharp edges, and rosin flesh shall be removed.

3.6 Workmanship. The pendulum, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, machine screw assemblage, welding, brazing, and freedom of parts from all burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject

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to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the pendulum shall be classified as follows:

(a) Preproduction inspection (paragraph 4.4)

(b) Quality conformance inspection (paragraph 4.5)

(c) Periodic production inspection (paragraph 4.4)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five pendulums, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the pendulum by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.5, a random sample of five pendulums will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The pendulums shall be

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considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.4.1 Vibration

4.4.1.1 Mounting. The pendulum, less pack or packaging, shall be mounted securely to a vibration device by means of its mounting holes and tested as specified in 4.4.1.2 and 4.4.1.3.

4.4.1.2 Vibration, nonoperating. The pendulum shall be subjected to the "Vibration, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.1.1.

4.4.1.3 Vibration, operating. The pendulum shall be subjected to the "Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.4.1.2.

4.4.2 Temperature and humidity

4.4.2.1 High temperature and humidity, nonoperating. The pendulum shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.1.

4.4.2.2 High temperature, operating. The pendulum shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.2.

4.4.2.3 Low temperature, nonoperating. The pendulum shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.4.2.3.

4.4.2.4 Low temperature, operating. The pendulum shall be subjected to the "Low temperatue, operating" test of MIL-E-82590 and, while at the test temperature, shall meet the requirements of 3.4.2.4.

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4.4.3 Operating life. Connect the pendulum as described in 4.5.4.1. Operate the tilt table alternately in each direction to 30 degrees plus or minus 10 degrees at frequency between one and 0.1 Hertz. After 500 hours the pendulum shall meet the performance requirements specified in 3.3.

4.5 Quality conformance inspection. Each pendulum delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.5.1 Test conditions. Unless otherwise specified, the pendulum shall be subjected to quality conformance tests under the following conditions:

(a) Temperature: 65 degrees F to 95 degrees F (18 degrees C to 35 degrees C).

(b) Altitude: Normal ground.

(c) Vibration: None

(d) Humidity: Room ambient to 95 percent relative maximum.

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities, and shall provide personnel for performing the quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.5.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.5.3.1 Mounting table, capable of being tilted from zero to 90 degrees in either direction and accurate within one minute of angle.

4.5.3.2 Vacuum tube ac voltmeter.

4.5.3.2.1 Range: 0.1, 0.3, 1, 3, 10, 30, 100, 300 volts.

4.5.3.2.2 Accuracy: 2 percent of full-scale reading.

4.5.3.2.3 Input impedance: 500,000 ohms minimum.

4.5.3.3 Megohmmeter.

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4.5.3.3.1 500 volts dc.

4.5.3.3.2 Capable of measuring 0 to 20 megohms.

4.5.3.4 Power sources.

4.5.3.4.1 115 volts plus or minus 3 percent, 400 Hz plus or minus 2 percent, single phase.

4.5.3.4.2 500 volts, 60 Hz (for hi-pot test).

4.5.3.5 Pinion Wrench, Drawing 677036.

4.5.3.6 Oscilloscope, Dumont type 304H or equivalent.

4.5.4 Test procedures

4.5.4.1 With the synchro pendulum completely assembled as per Drawing 877366, except for the cover, fasten the base to a tilt table so that the pendulum hangs down. The tilting axis of the table must be parallel to the pivoting axis of the pendulum.

4.5.4.2 Damping. With the table and pendulum level, deflect the pendulum by hand as far as possible. Suddenly release the pendulum and allow it to swing to rest naturally. The pendulum shall come to rest within six cycles to meet the requirements of 3.3.7.

4.5.4.3 Install the cover. Connect the ac voltmeter across the synchro output leads and apply 115 volts rms, ac, 400 Hz across pins 1 and 2 of connector P1, with phase A connected to pin 1 and phase B connected to pin 2. With the table and pendulum level, measure the output voltage. It shall not exceed 0.13 volt rms.

4.5.4.3.1 Deflect the pendulum by hand in each direction until the edge of the pendulum is even with the edge of the magnet (Drawing 859875). Suddenly release the pendulum and allow it to swing to rest. The two voltages (one from each direction) after the pendulum returns to its rest position and the voltage in 4.5.4.3 must be equal within 0.05 volt as specified in 3.3.2.

4.5.4.4 Tilt the mounting table slowly in each direction and observe the voltage change. The voltage shall vary smoothly as described in 3.3.3.

4.5.4.5 Measure the output voltage for tilt angles ranging from zero to ten degrees in one degree steps in each direction. The output voltage shall conform to the requirements in 3.3.4 and 3.3.6.

4.5.4.6 Tilt the synchro pendulum ninety degrees from level in each

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direction in five-degree steps and measure the voltages produced. The voltage shall agree as in 3.3.6.

4.5.4.7 Phase. With the tilt stand level and pin 1 connected to the X input, pin 7 connected to the Y input, and pins 2 and 8 connected to ground of the oscilloscope, apply 115 volts rms ac 400 Hz to pins 1 and 2. Move the tilt stand counter clockwise (when viewed from the pendulum end). The Lissajous figure shall rotate in the same direction to verify the requirements of 3.3.5.

4.5.4.8 Movement. The pendulum shall swing freely and shall not touch the cover nor any part of the framework when the pendulum base is rotated in either direction 90 degrees from level position.

4.5.4.9 Insulation. With the atmospheric conditions as specified in 4.5.1, make the following tests:

4.5.4.9.1 Connect pins 1, 2, 7, and 8 of P1 together.

4.5.4.9.2 A megohmmeter reading between pins 1, 2, 7, and 8 and ground shall be not less than 20 megohms at 500 volts dc.

4.5.4.9.3 Apply 500 volts, 60 Hz between pins 1, 2, 7, and 8 and ground for one minute. There shall be no evidence of breakdown.

4.5.4.9.4 Apply 500 volts, 60 Hz between pins 1 and 2 connected together and pins 7 and 8 connected together. There shall be no evidence of breakdown.

4.5.4.10 Visual and dimensional examination. Each pendulum shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.5.4.11 Preservation, packaging, packing, and marking. The pendulum, when ready for delivery, shall be inspected for compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and packaging shall be as specified (see 6.2)

5.1.1 Level A

5.1.1.1 Unit packaging. The pendulum shall be packaged in accordance with Method IID of MIL-P-116 and LD 495859.

5.1.2 Level B. Not applicable.

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5.1.3 Level C

5.1.3.1 Unit packaging. The pendulum shall be packaged in accordance with Method III of MIL-P-116.

5.2 Packing. Packing shall be level A or Level C, as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be as specified in the contract or order.

5.2.2 Level A

5.2.2.1 Unit pack. The pendulum shall be packed in accordance with LD 495859 and may be shipped individually in small lots.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Load Type III (see 6.2).

5.2.3 Level B. Not applicable.

5.2.4 Level C. The pendulum shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Normal markings Level A. Unit packs shall be marked in accordance with LD 495859.

5.3.2 Normal markings Level C. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.3 Serial numbers. Unit packages and shipping containers shall be marked with the serial numbers of the units contained within.

6. NOTES

6.1 Intended use. The pendulum is intended for use in Torpedo Mk 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

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(d) Required level of packing, including palletizing (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.3 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

6.3.1 Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

6.3.2 Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

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